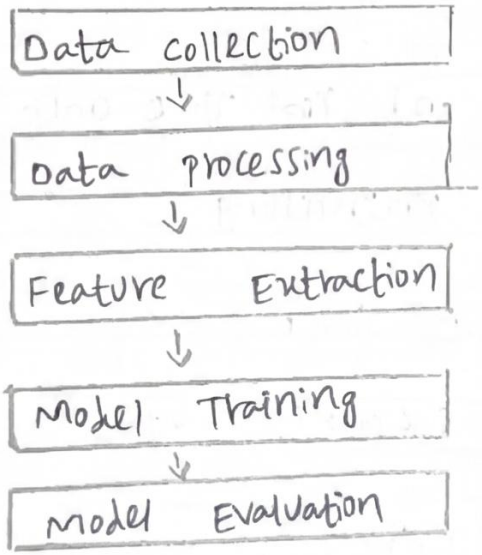
Kadhirezhilan D
192511228
ITA 0612

CO2-AT2 - Scenario based questions

1. Functional Blocks of a Machine Learning system

->A machine learning (ML) system learns from data and uses the learned patterns to make predictions or find groups

Flow chart:



Types of learning:

-> supervised learning : used labelled data

ex: predicting wheather

-> un-supervised learning : uses un-labelled data to find patterns

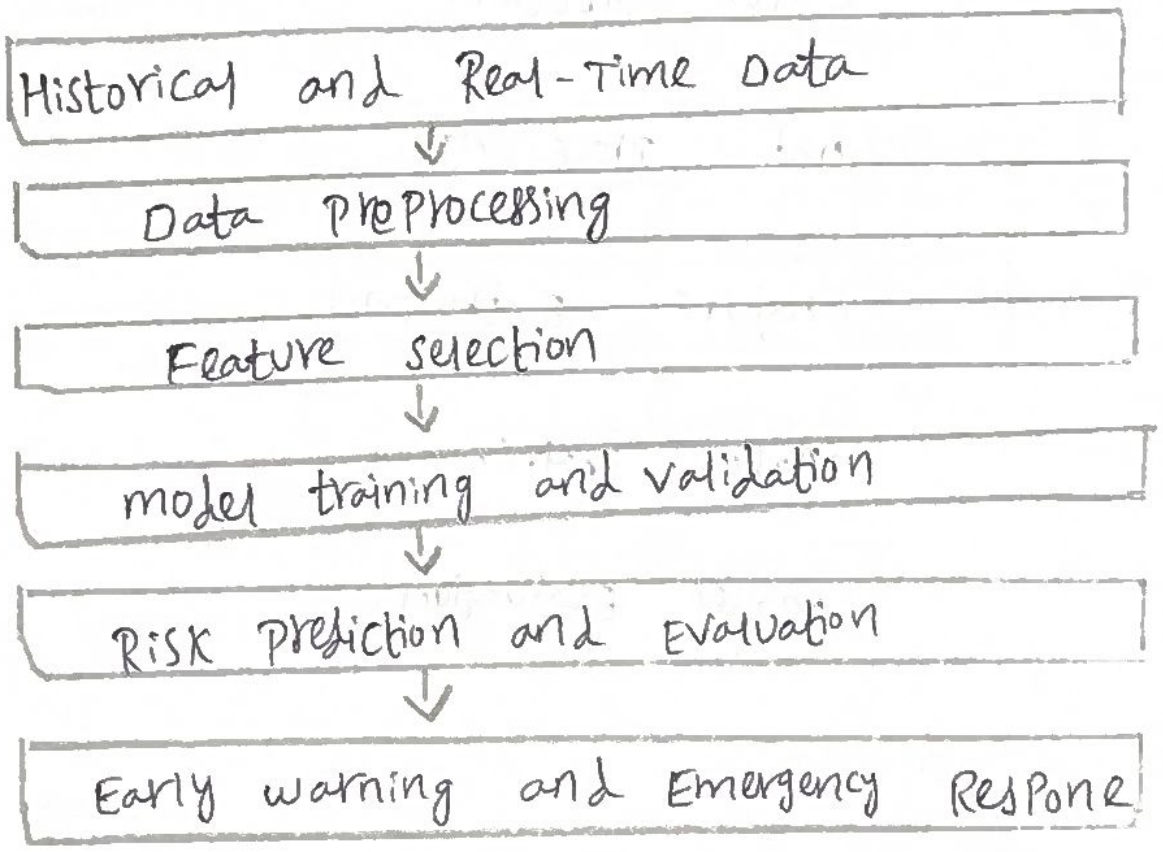
ex :- grouping customers based on their purchasing habits.

Conclusion : These blocks help a machine learning system learn from data and produce useful results.

2. Machine Learning system for Disaster management

→ A machine learning system can help predict disaster risks, such as floods and earthquakes, using historical and real time data

Flow Chart :

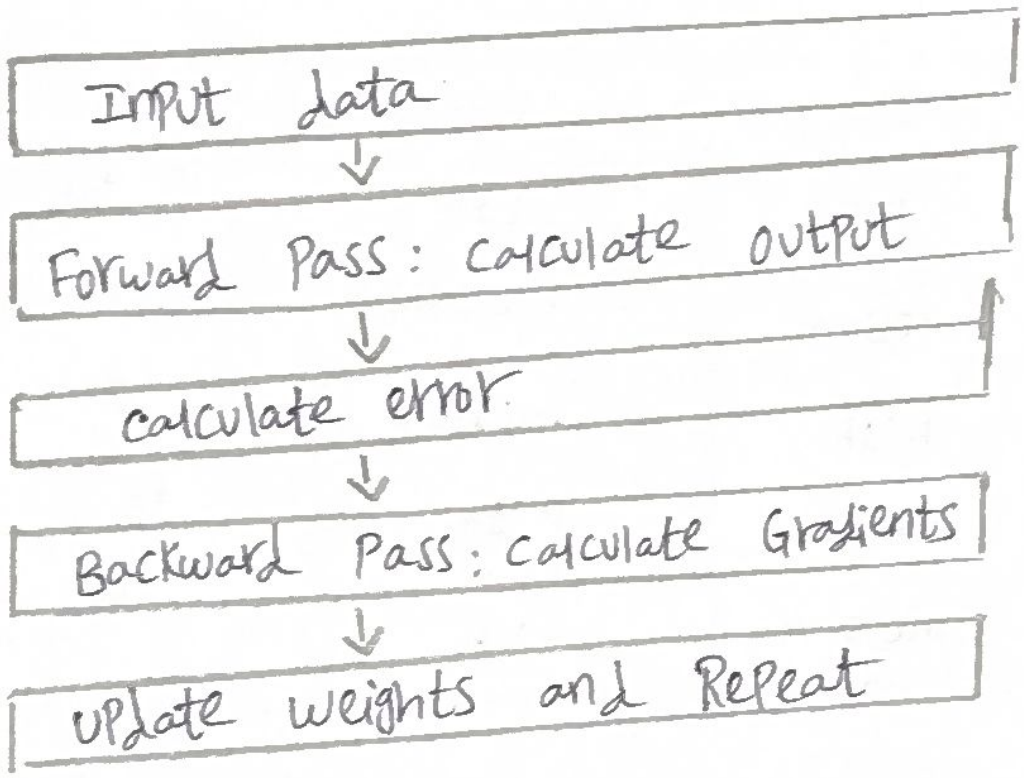


CONCLUSION : The system helps authorities assess disaster risks and prepare emergency responses. earthquakes are difficult to predict precisely, so the system should focus on scientifically supported risk estimates and early-warning signals rather than claiming exact predictions.

3. Back Propagation in a multilayer Perceptron (MLP)

→ Back Propagation is learning algorithm that adjusts neural network weights to reduce Prediction errors.

Flow Chart :



The basic update weight rule:

$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$$

Ex : If the target output is 1 but the model predicts 0.6, the model has made an error

CONCLUSION : BackPropagation improves the network by repeatedly updating weights and reducing prediction errors.

4. Genetic Algorithm - Speed and Accuracy

→ A Genetic Algorithm (GA) finds good solution by imitating natural selection and evolution

The given data set:

S.No	Speed	Accuracy	Fitness	suitable
1	High	Low	40	NO
2	medium	medium	60	Yes
3	High	medium	75	Yes
4	Low	High	65	Yes
5	medium	High	80	Yes

step -1 : selection

→ The best candidate is example 5, with fitness 80. other suitable candidates include Examples 3, 4 & 2.

step-2 : crossover

→ cross over combines Parts of two selected Parents to generate new off spring.

step-3 : Mutation

→ Mutation randomly changes Part of a candidate solution. For example, it may change a medium - speed feature to high speed.

Step-4: Evolution across generations

→ Final selection: Keep the candidate with the highest evaluated fitness.

5. Neural Network for Disease classification

→ A Neural network can predict whether a Patient has a disease using age, blood pressure, glucose level, and lifestyle risk.

calculate matrix:

S.NO	Actual output	Predicted output	Result
1	Yes	Yes	TP
2	Yes	NO	FN
3	NO	NO	TN
4	Yes	Yes	TP
5	NO	Yes	FP

∴ TP = 2

TN = 1

FP = 1

FN = 1

Calculate evaluation metrics

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \times 100$$

$$= \frac{2+1}{5} \times 100$$

$$= 60\%$$

$$\text{Precision} = \frac{TP}{TP+FP} \times 100$$

$$= \frac{2}{3} \times 100$$

$$= 66.67\%$$

$$\text{Recall} = \frac{TP}{TP+FN} \times 100$$

$$= \frac{2}{3} \times 100$$

$$= 66.67\%$$

$$\text{Specificity} = \frac{TN}{TN+FP} \times 100$$

$$= \frac{1}{2} \times 100$$

$$= 50\%$$